Solution: Subsets

Problem Statement

Given a set with distinct elements, find all of its distinct subsets.

Example 1:

```
Input: [1, 3]
Output: [], [1], [3], [1,3]
```

Example 2:

```
Input: [1, 5, 3]
Output: [], [1], [5], [3], [1,5], [1,3], [5,3], [1,5,3]
```

Solution

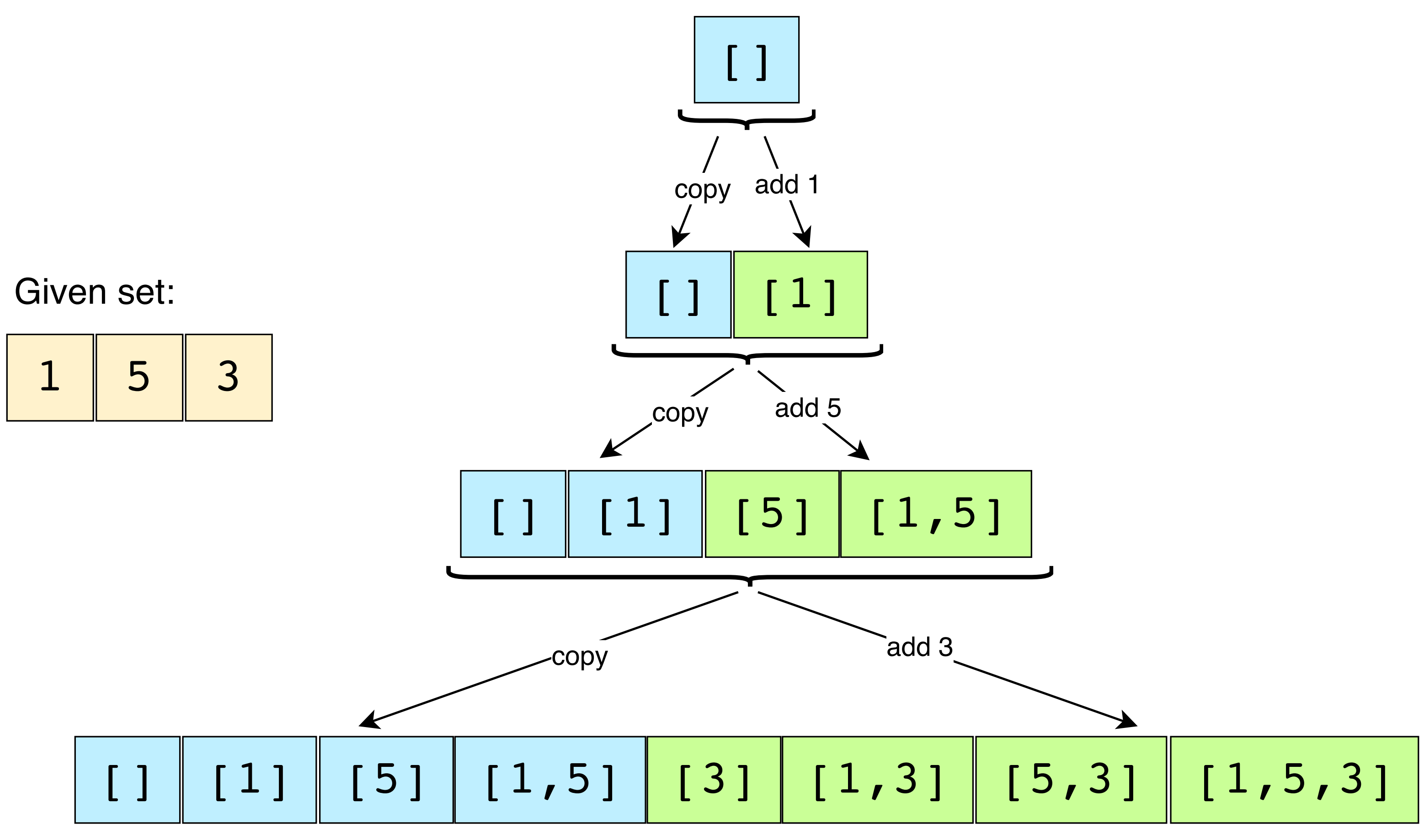
To generate all subsets of the given set, we can use the **Breadth First Search** (**BFS**) approach. We can start with an empty set, iterate through all numbers one-by-one, and add them to existing sets to create new subsets.

Let's take the example-2 mentioned above to go through each step of our algorithm:

Given set: [1, 5, 3]

1. Start with an empty set: [[]]
2. Add the first number (1) to all the existing subsets to create new subsets: [[], [1]];
3. Add the second number (5) to all the existing subsets: [[], [1], [5], [1,5]];
4. Add the third number (3) to all the existing subsets: [[], [1], [5], [1,5], [3], [1,3], [5,3], [1,5,3]].

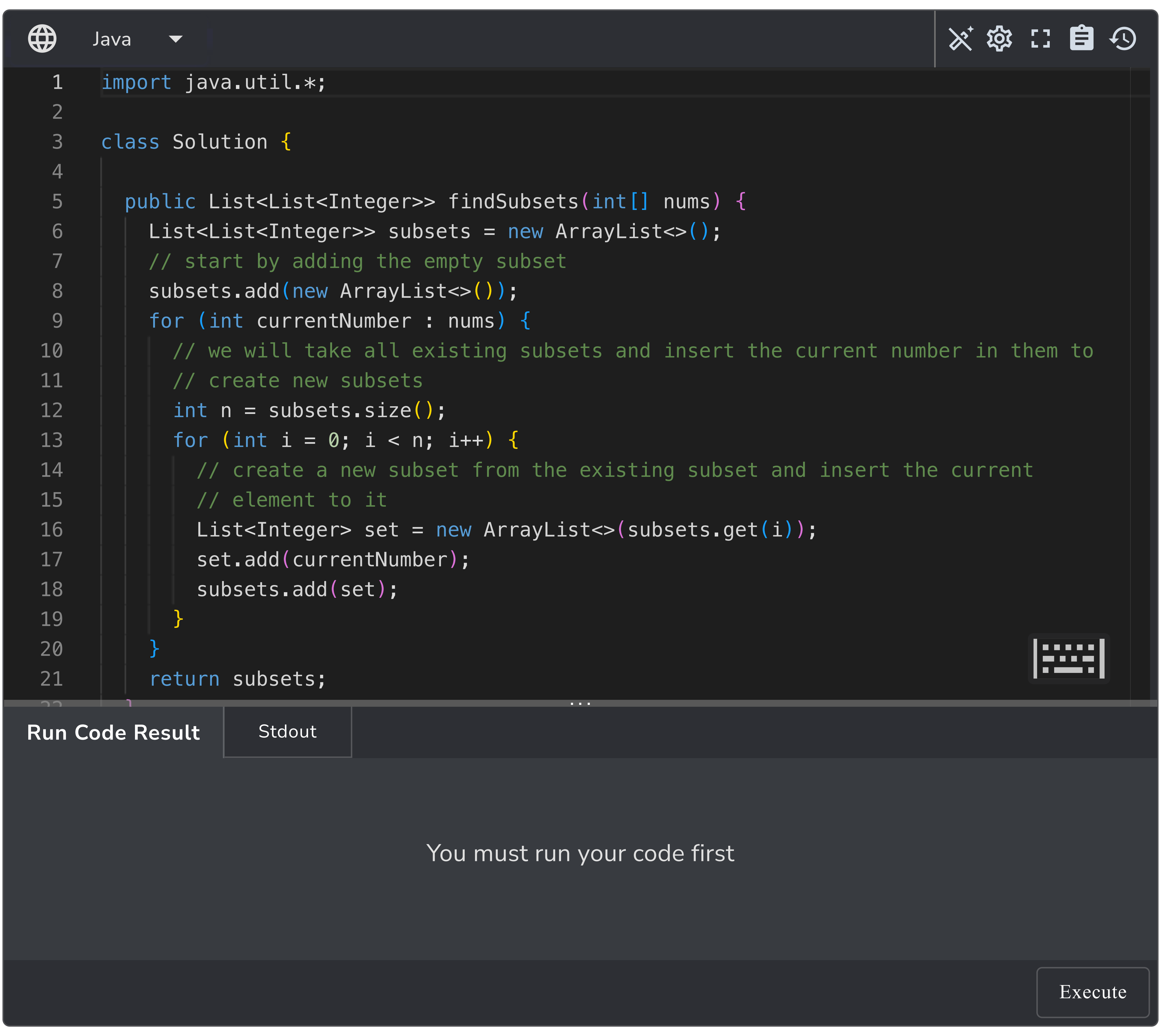
Here is the visual representation of the above steps:



Since the input set has distinct elements, the above steps will ensure that we will not have any duplicate subsets.

Code

Here is what our algorithm will look like:



```java
import java.util.*;

class Solution {

  public List<List<Integer>> findSubsets(int[] nums) {
    List<List<Integer>> subsets = new ArrayList<>();
    // start by adding the empty subset
    subsets.add(new ArrayList<>());
    for (int currentNumber : nums) {
      // we will take all existing subsets and insert the current number in them to
      // create new subsets
      int n = subsets.size();
      for (int i = 0; i < n; i++) {
        // create a new subset from the existing subset and insert the current
        // element to it
        List<Integer> set = new ArrayList<>(subsets.get(i));
        set.add(currentNumber);
        subsets.add(set);
      }
    }
    return subsets;
```

You must run your code first

Time Complexity

Since, in each step, the number of subsets doubles as we add each element to all the existing subsets, therefore, we will have a total of $O(2^N)$ subsets, where 'N' is the total number of elements in the input set. And since we construct a new subset from an existing set, therefore, the time complexity of the above algorithm will be $O(N*2^N)$.

Space Complexity

All the additional space used by our algorithm is for the output list. Since we will have a total of $O(2^N)$subsets, and each subset can take up to O(N) space, therefore, the space complexity of our algorithm will be $O(N*2^N)$.

← Previous
Subsets (Easy)

Next →
Subsets With Duplicates (Easy)

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